Module-1: Introduction and Basic Definitions

Thermodynamics (NMEC103) : (Winter 2024-25)

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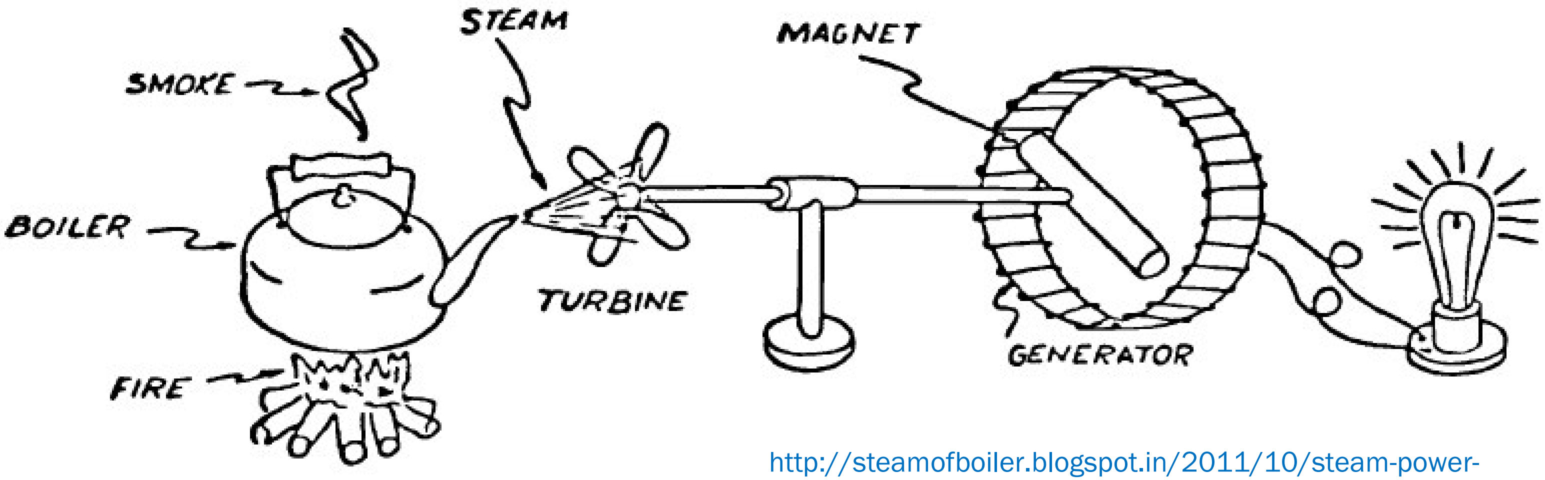
Introduction

Thermodynamics is the branch of science which deals with energy, its transfer, and transformation. The science that deals with matter, energy, and interaction between matter and energy

Thermodynamics is derived from two Greek words

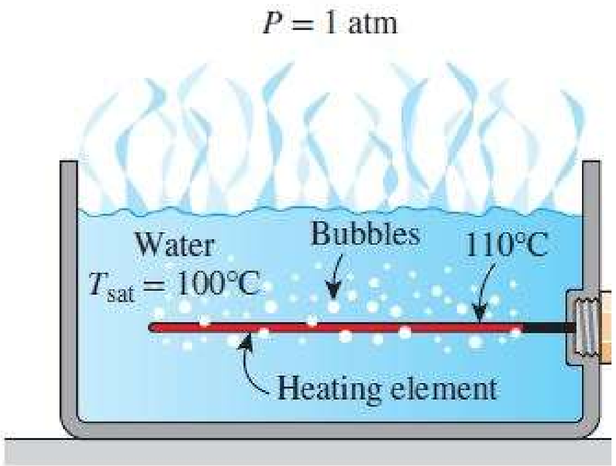
Therme+ Dynamics

Heat   Force ➡ motion ➡ power

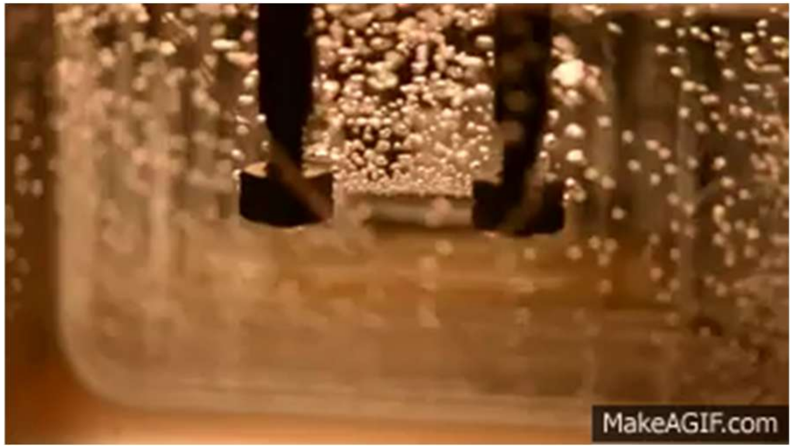


http://steamofboiler.blogspot.in/2011/10/steam-power-plant.html

The term “dynamics ” in thermodynamics refers to the dynamics of the mechanical motion associated with the process of converting heat into work. It does not refer to the dynamics of nonequilibrium thermodynamic states as the subject of Classical Thermodynamics is based on the study of equilibrium states




(a)



Thermodynamics can not answer the dynamics associated with the highly non-equilibrium boiling process

- One definition of thermodynamics is that it is a science that deals with heat and work and those properties that bear a relationship with heat and work (Classical definition)
- However modern thermodynamics deals with more than just thermal energy. Hence it is more appropriate to define thermodynamics as:

"The science and technology that deals with the laws that govern the transformation of energy from one form to another" (Modern definition)

Like all sciences, thermodynamics has its basis in experimental observation. The observations have been formulated into certain basic laws, which are applied along with the relevant properties to various energy conversion systems and processes, with a view to design, develop and analyse these systems and processes.

- Applications:

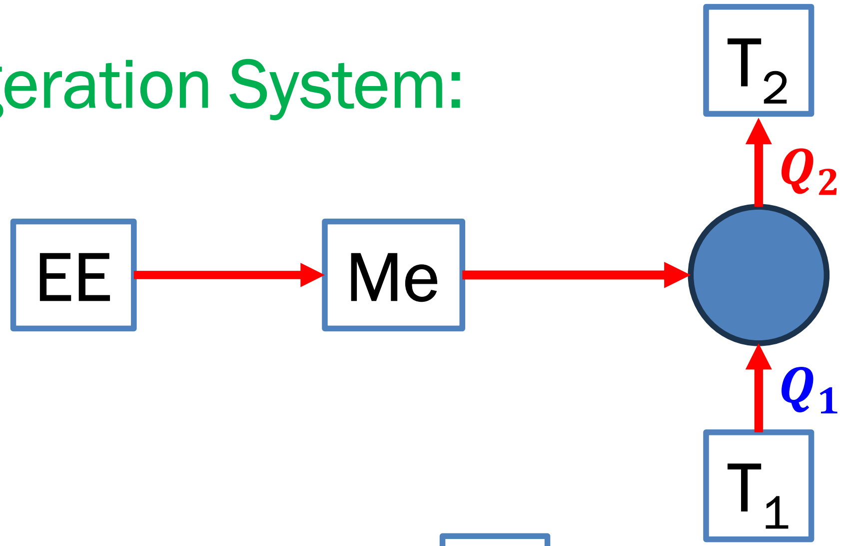
- Since thermodynamics is a science that deals with energy transformations, its application is global in nature

- Thermodynamics uses the powerful "black box " approach for the analysis, as a result it can be applied to the smallest component of a system or to the whole complicated system itself.

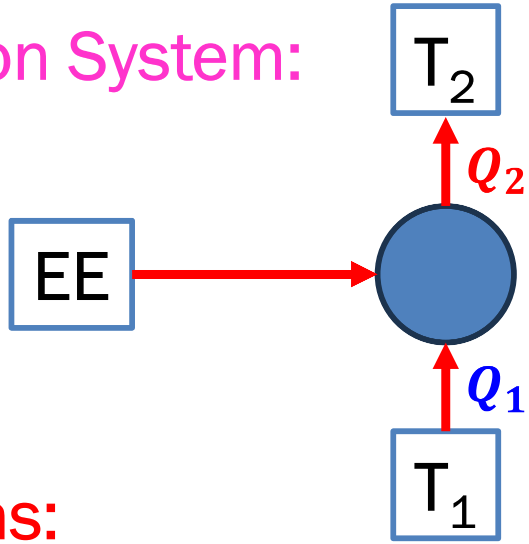
- Thermal Power plants: Ch → Th → Me → EE
- Fuel Cells: Ch → EE

- Power Producing Systems
  - I. C. Engines
  - Gas Turbines
  - Nuclear Power Plants
  - Chemical Rocket Engines

- Vapor Compression Refrigeration System:



- Thermoelectric Refrigeration System:



- Miscellaneous Applications:
  - Cryogenics (Gas liquefaction, Air separation)
  - Chemical Process
  - Metallurgical Process
  - Biological Thermodynamics

- Metabolism

For warm blooded animals (Mice $\longrightarrow$ Elephant)

$$BMR = 293M^{0.75}$$ Basal Metabolic Rate

BMR in kJ/day, M is the body mass in kg

$$\frac{BMR}{M} = 293M^{-0.25} \Longrightarrow \frac{BMR}{M} \text{ increases } as\ M \text{ decreses}$$

As M decreases (A/V) ratio increases $\Longrightarrow$ more heat loss $\Longrightarrow$ higher metabolic rate required to keep the body warmer $\Longrightarrow$ More frequent feeding required for smaller animals otherwise they will starve $\Longrightarrow$ There is a lower limit to the size of the warm blooded animals (humming bird) Thermodynamic Lower Limit of Body Size

- Nutrition & Diet

$$\eta_{Carbohydrate} = \frac{E_{out}}{E_{in}} = \frac{17.2}{18.0} = 95.5\%$$

$$\eta_{Protein} = \frac{E_{out}}{E_{in}} = \frac{17.2}{22.0} = 77.5\%$$

$$\eta_{fat} = \frac{E_{out}}{E_{in}} = \frac{38.9}{39.8} = 97.5\%$$

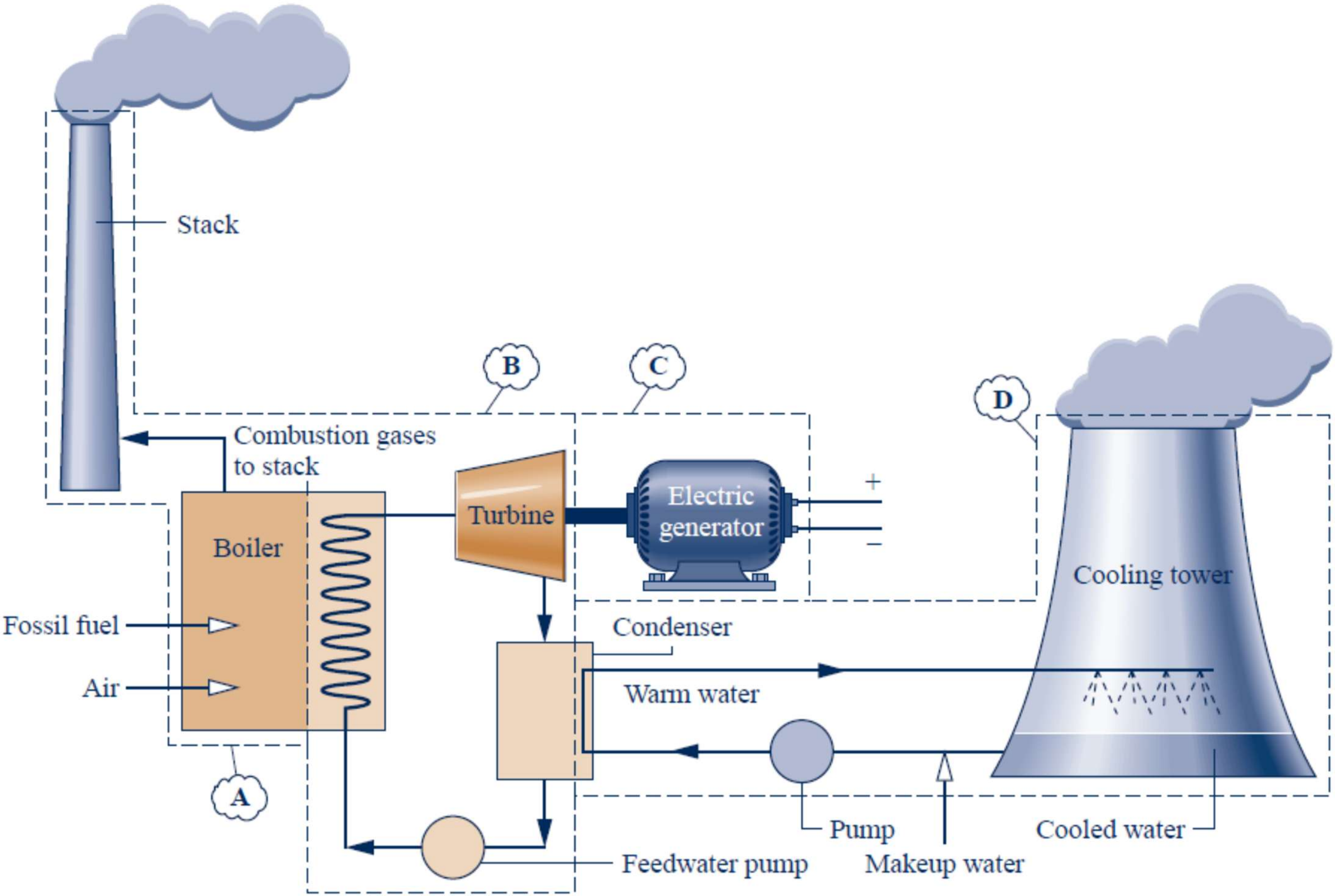
Over weight: Each kg of body tissue consists of 0.885 km of tiny blood vessel. Thus, if a person is overweight by 10 kg $\Longrightarrow$ Heart must pump blood through an extra $0.885 \times 10 = 8.85$ km of blood vessel

- 10 kg of fat can sustain 40 days without food $\Longrightarrow$ explains why dieting is a very slow weight loss process

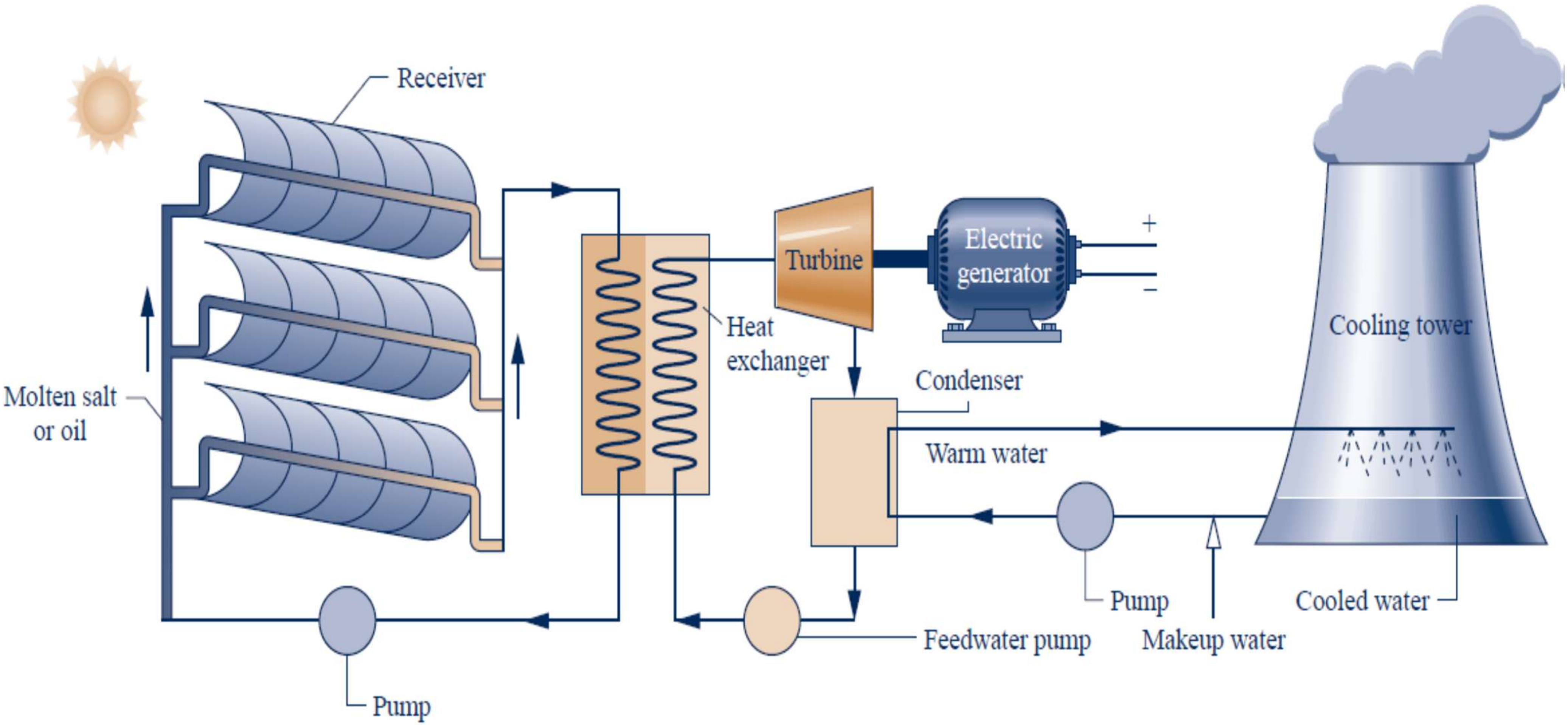
- However, during fasting body also consumes proteins in body $\Longrightarrow$ dangerous

- Man is known to have survived w/o food for 100 days however a man can not survive w/o water for more than 10-20 days (continuous water loss due to respiration and perspiration)

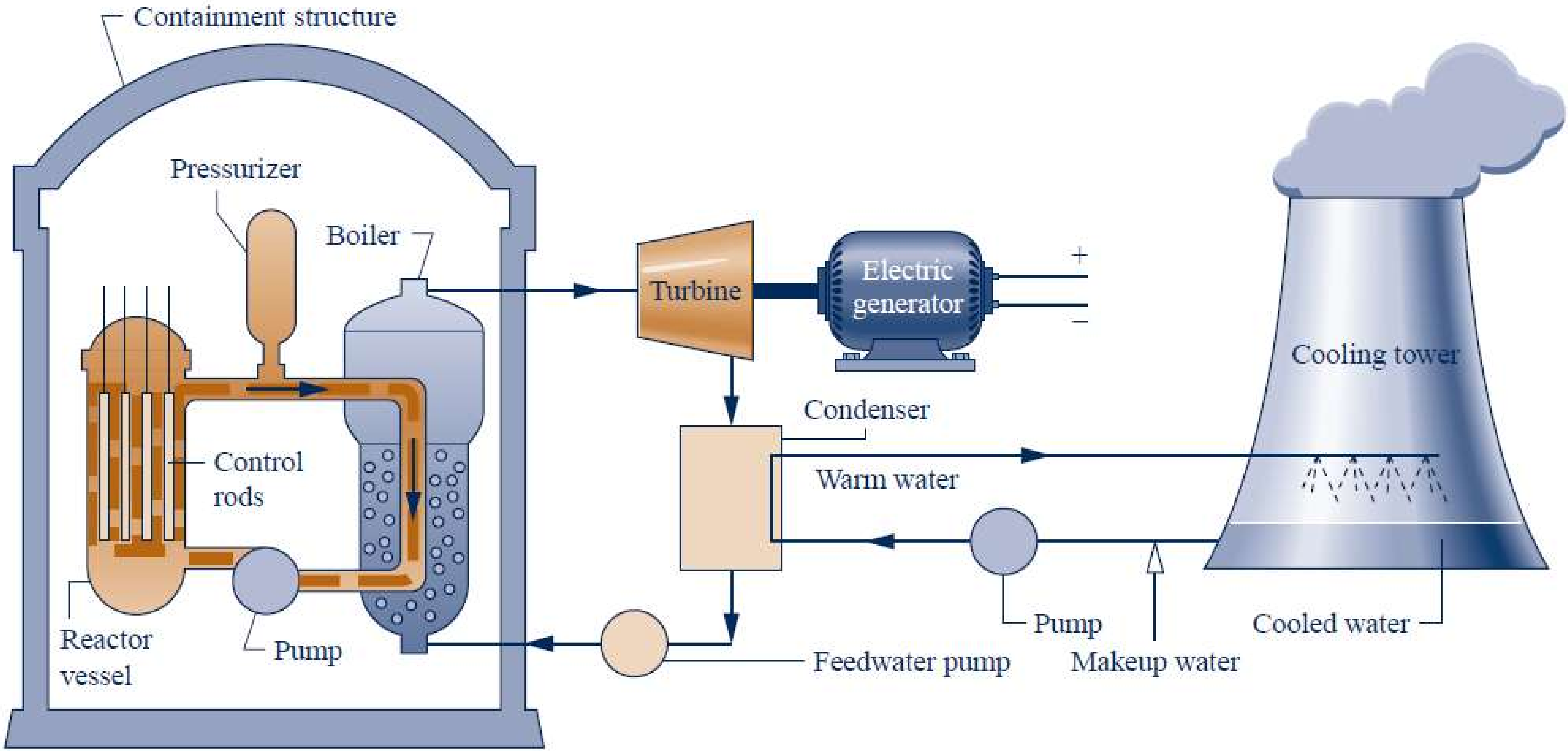
- Physiologically it is very hard to lose weight, the apparent weight loss after exercise is due to the water loss by perspiration



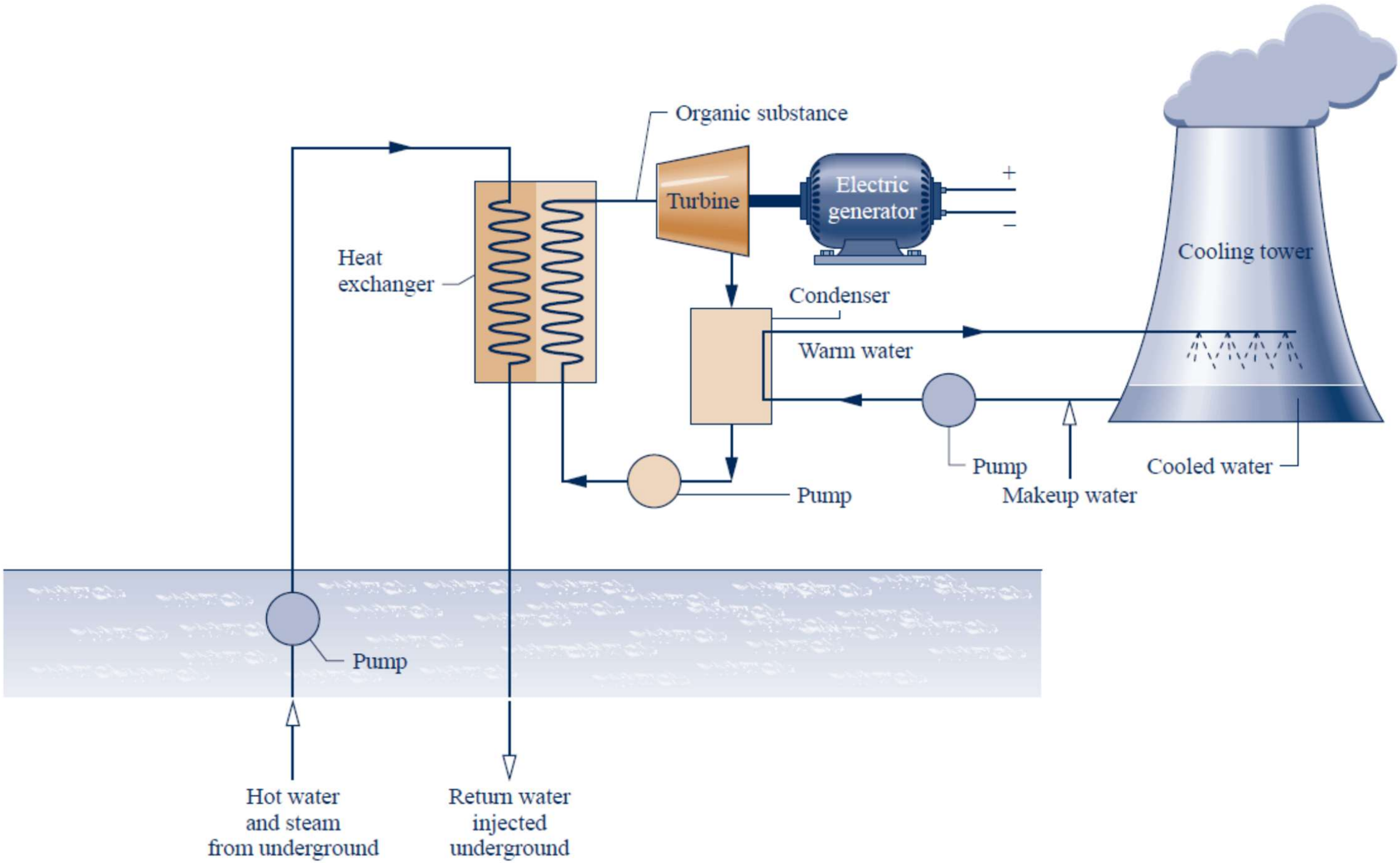
(*a*) Fossil-fueled vapor power plant.



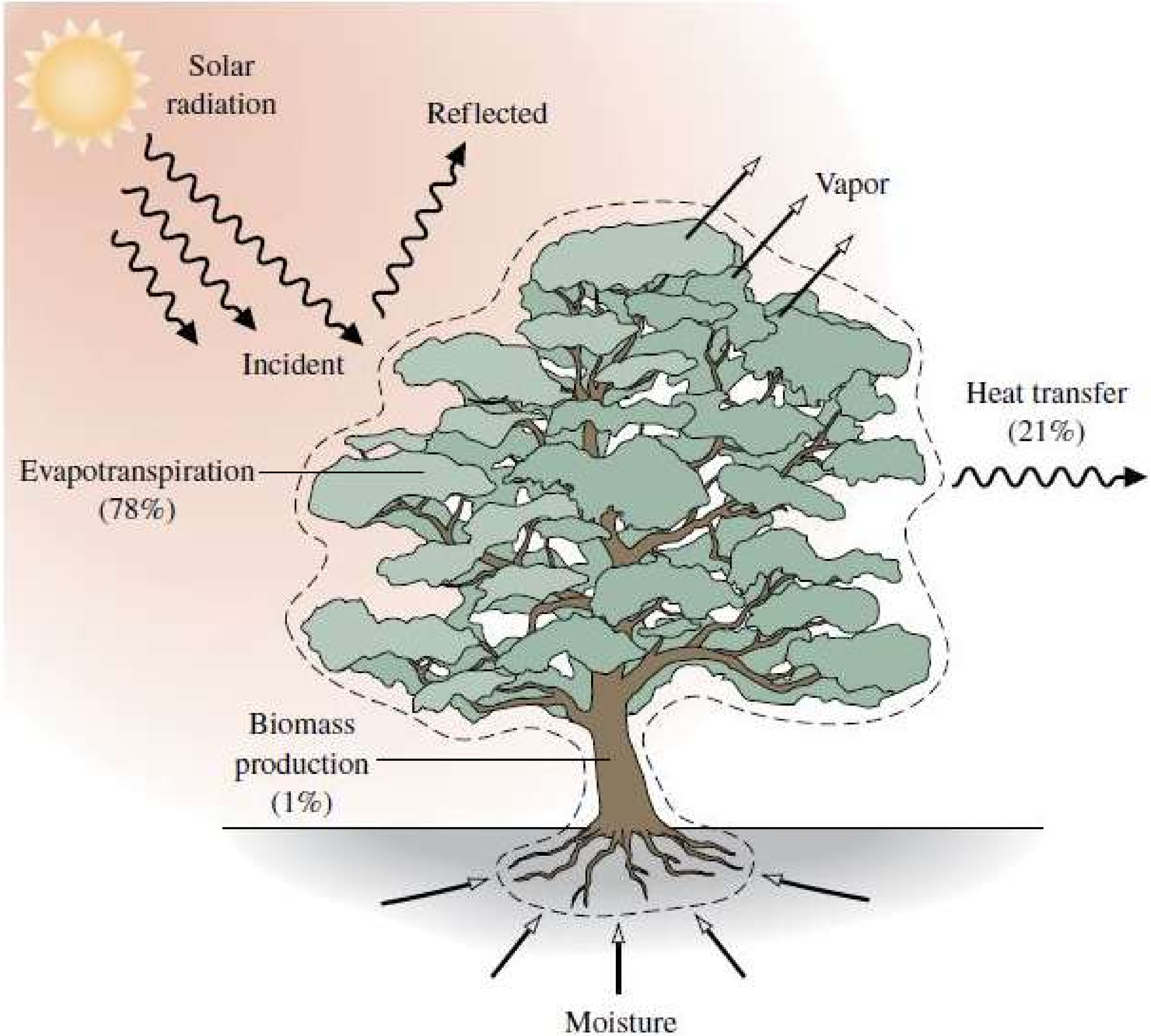
(*c*) Concentrating solar thermal vapor power plant.



(*b*) Pressurized-water reactor nuclear vapor power plant.



(*d*) Geothermal vapor power plant.


Solar
radiation
Reflected
Vapor
Incident
Heat transfer
(21%)
Evapotranspiration
(78%)
Biomass
production
(1%)
Moisture

Some application areas of thermodynamics.

Refrigerator

Boats

© *Doug Menuez/Getty Images RF*

Aircraft and spacecraft

© *PhotoLink/Getty Images RF*

Power plants

© *Malcolm Fife/Getty Images RF*

Human body

© *Ryan McVay/Getty Images RF*

Cars

© *Mark Evans/Getty Images RF*

Wind turbines

© *F. Schussler/PhotoLink/Getty Images RF*

Food processing

Glow Images RF

A piping network in an industrial facility.

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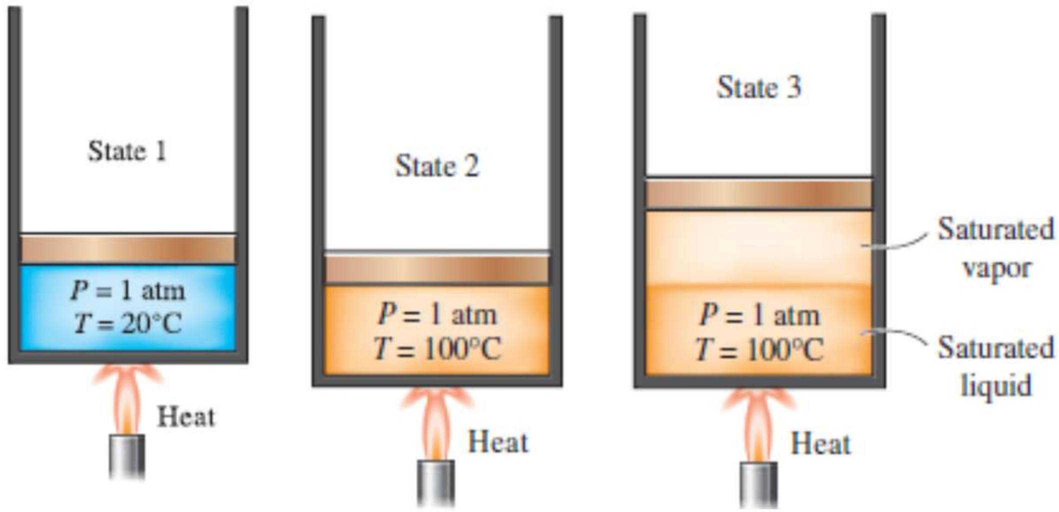
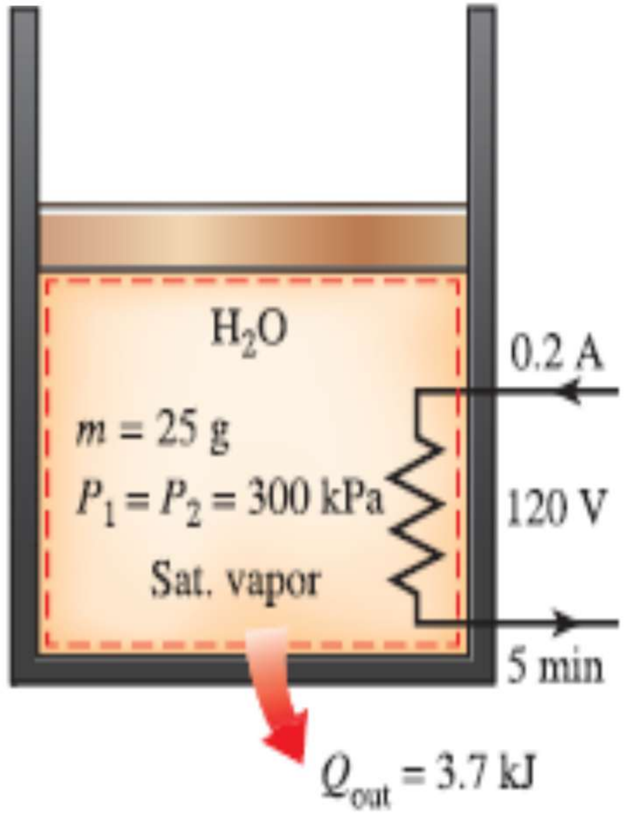
Introduction (Four Es)

Thermodynamics tells us how to convert heat into power

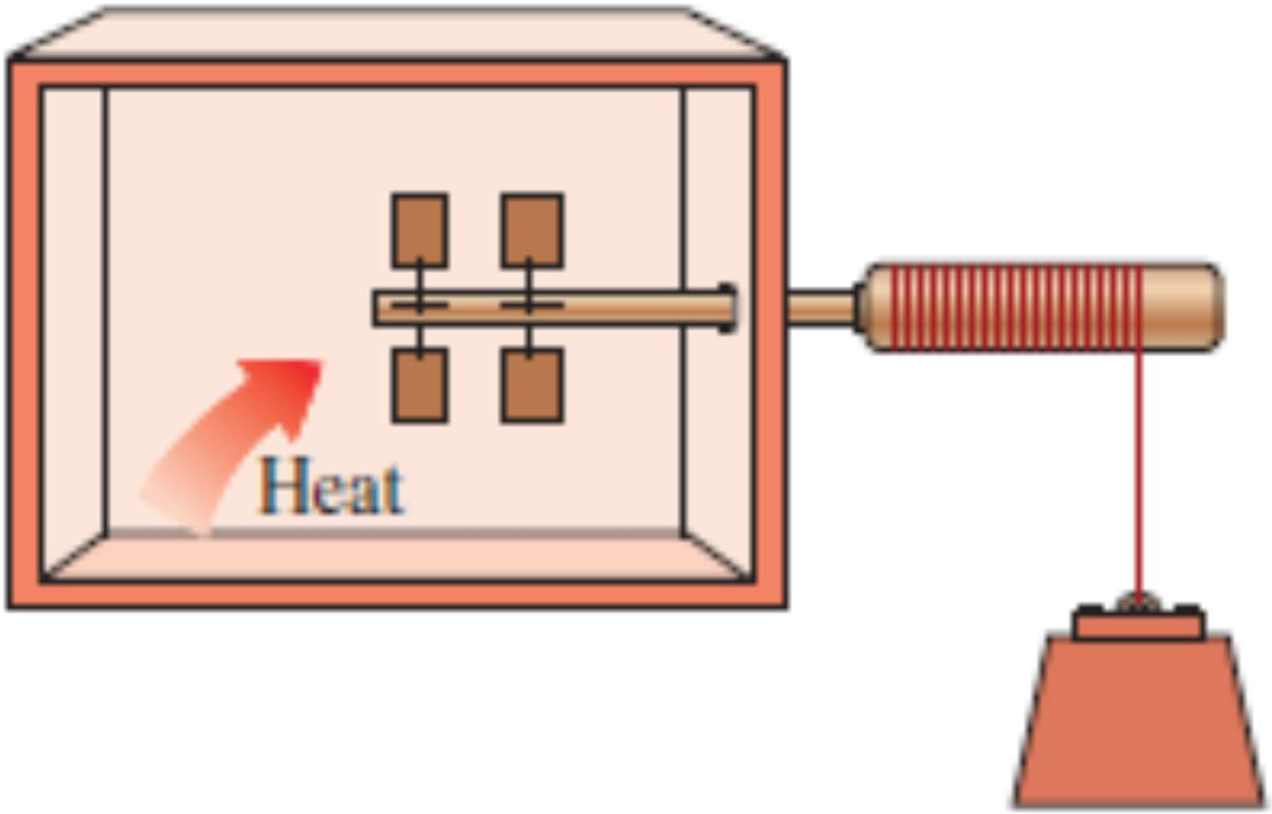
a) Thermodynamics deals with Equilibrium states, and is able to answer that "(a) how much Energy interaction is required to bring a system from one Equilibrium state to other ".



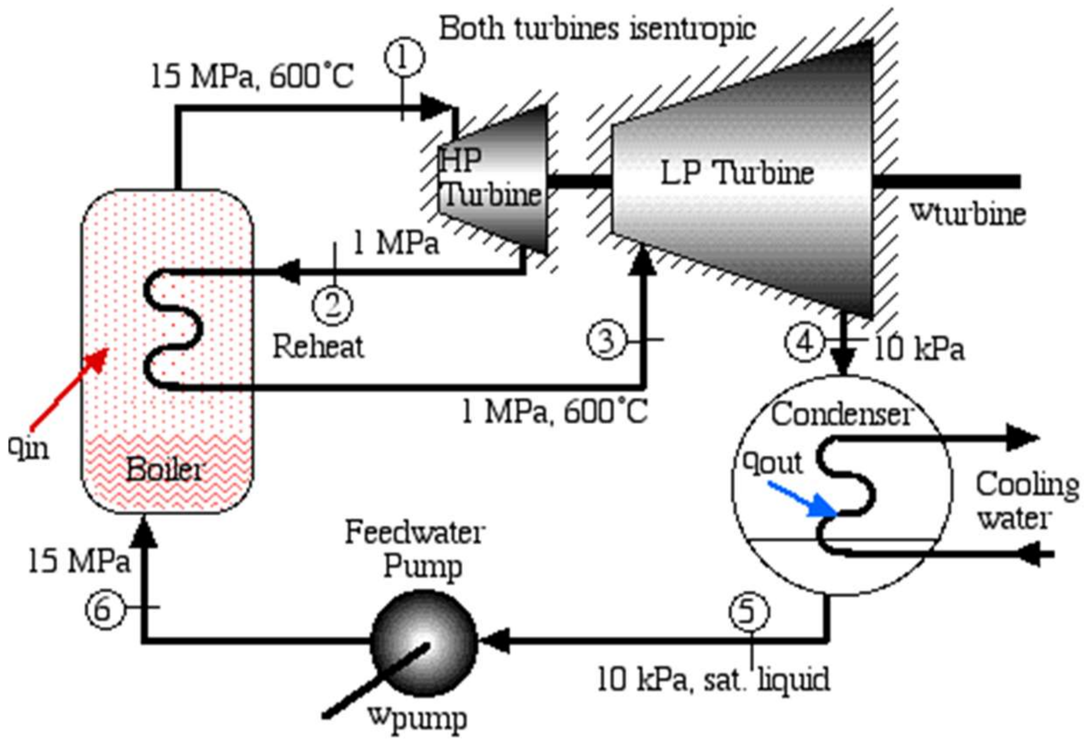
(b) On the other hand, if the energy interactions are known, Thermodynamics is also capable of specifying the final Equilibrium state. (A state which every isolated system with no internal constraints will eventually attains)



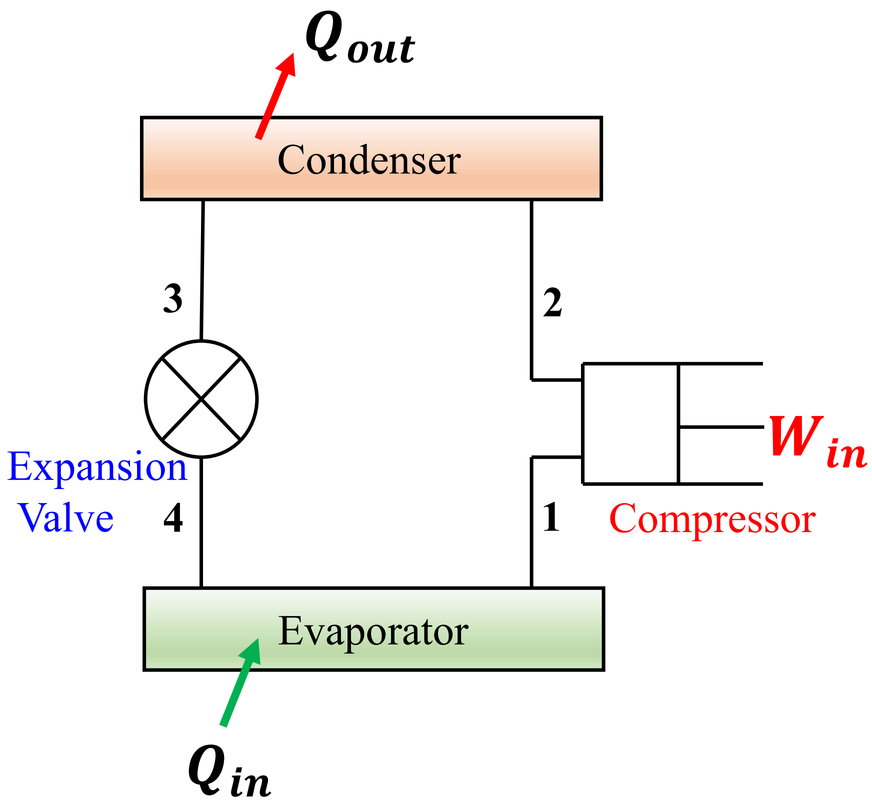
(c) Thermodynamics is also equipped with a powerful tool called Entropy by which it is capable of discarding impossible processes and identifying the possible direction as well as limit of any process. No change which results in a decrease in entropy of the universe can occur.



d) Every subject has a keyword and for engineering it is Efficiency. Thermodynamics defines, efficiency of various work producing (Heat engine) and work consuming machines



Heat engine



Refrigerator

$$\eta = \frac{W_{net}}{Q_{in}}$$

$$COP = \frac{Q_{in}}{W_{in}}$$

Mechanics & Thermodynamics

In mechanics, before analysing any problem we draw the free body diagram, show all the active and reactive forces acting on the free body, apply the conditions of equilibrium or laws of motion.

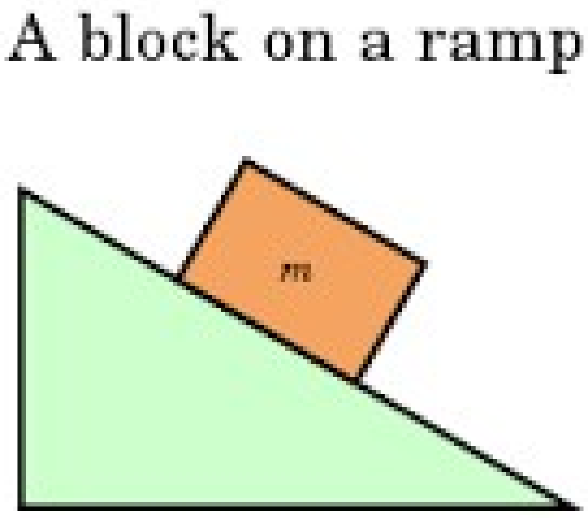
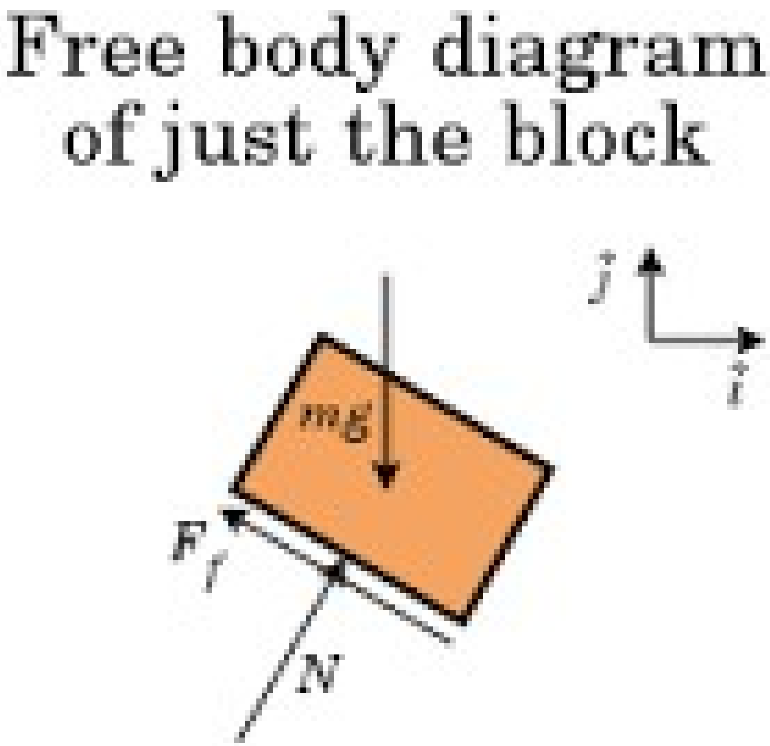
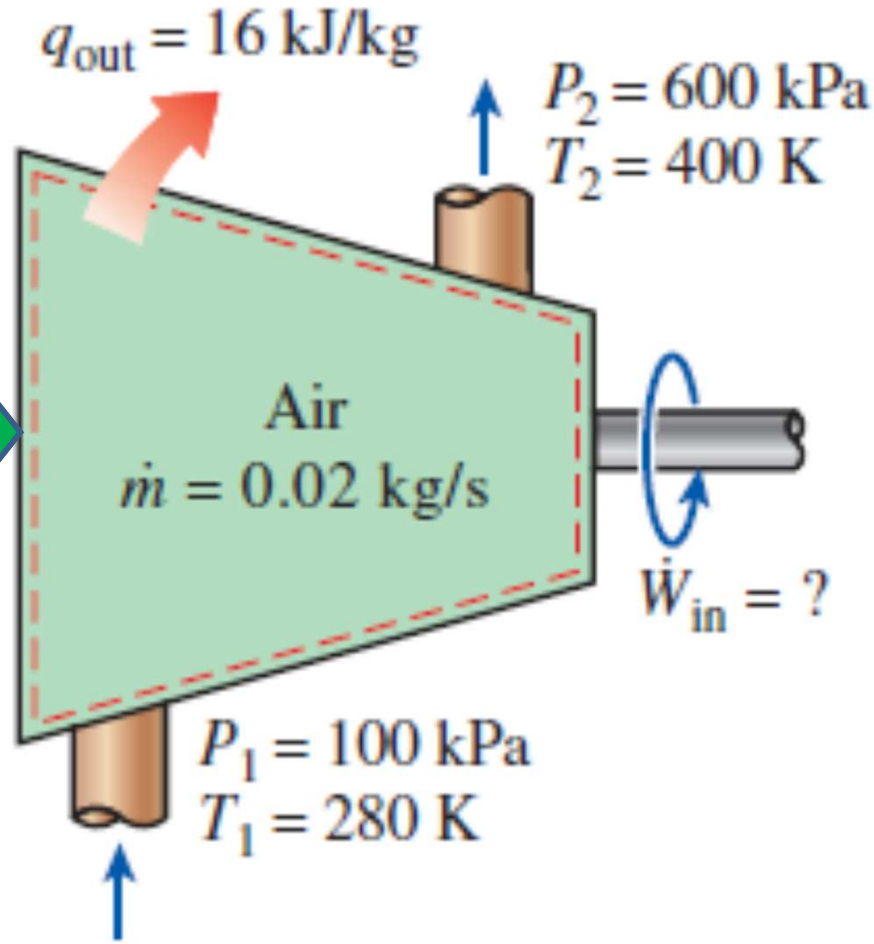


Figure shows the free body diagram of the block. All the active forces (weight and friction) reactive force (N) are shown



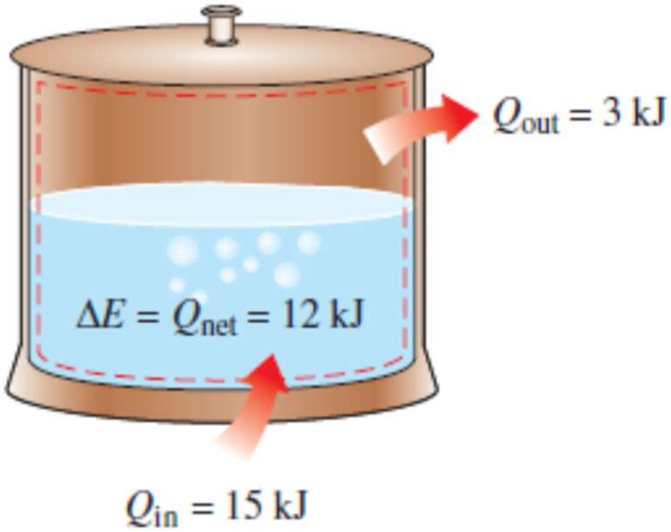
In Thermodynamics before analysing any problem we identify the system/control volume, show all the energy & mass interactions to or from the system as well as specify the initial and or final state of the system, apply the laws of Thermodynamics

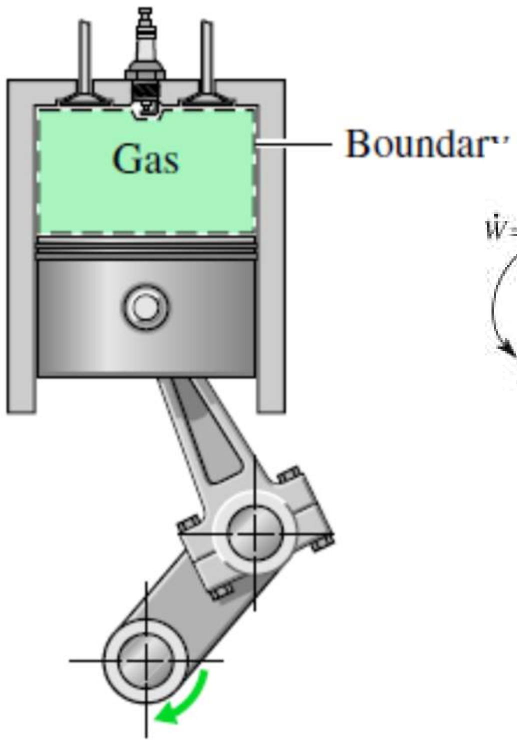
Figure shows the Compressor as the control volume, all the energy and mass interaction with inlet and exit conditions are specified

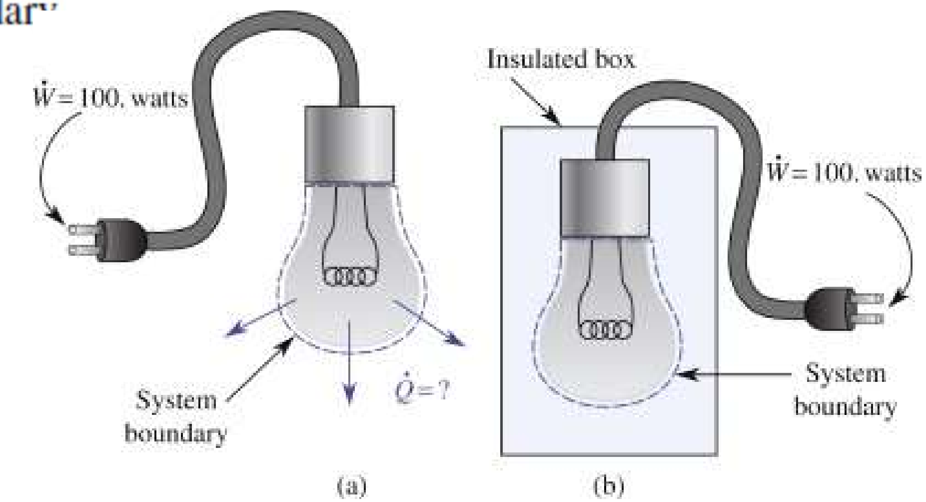


Thermodynamic System (Control Mass/ Closed system)

A thermodynamic system (Control Mass) is defined as a quantity of matter of fixed mass and identity upon which attention is focused for study. Often, the system is only a collection of particles of matter (such as a number of electrons, protons, atoms or molecules). Sometimes, the system may contain matter and radiation (photons) or even radiation only. In simple terms, a system is whatever we want to study. The boundary of control mass/ closed system is not permeable for mass transfer. The boundary may be rigid/ movable. The system as a whole may be stationary or movable with respect to an external frame of reference. A closed system is a constant mass system but all constant mass systems are not closed in nature. The boundary of a closed system can not be imaginary.

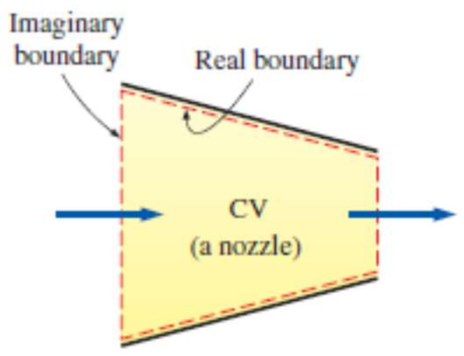




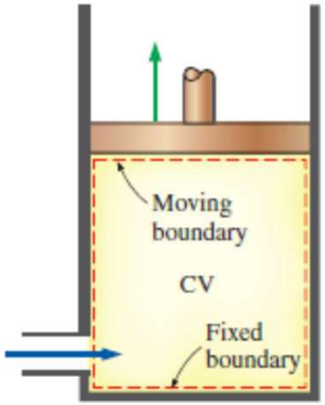


Control Volume (Open System)

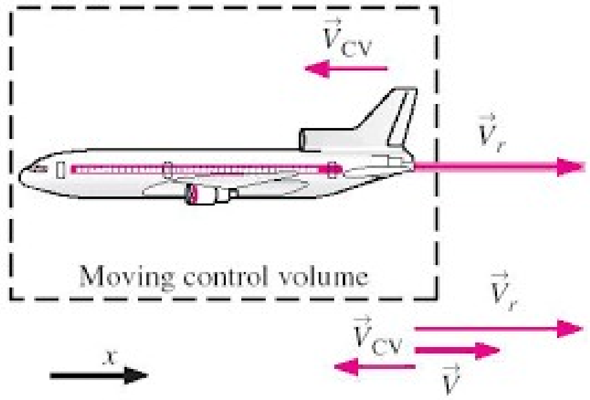
A control volume (does not mean that volume is fixed ,i.e. we may have a control volume with moving /deformable boundary (transient case)), which may be considered as an open system, is a specified region in space upon which attention is focused. The control volume is separated from the surroundings by a control surface (may be real/imaginary) which may be permeable for mass and energy transfer. The control volume may be fixed may be movable. The Control Volume may be stationary or movable.



(b) A control volume with real and imaginary boundary

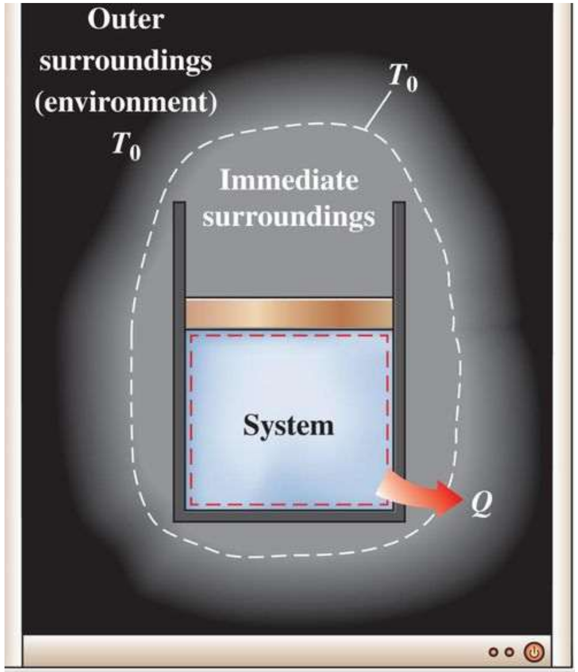


(b) A control volume with moving boundary (deformable control volume)



(c) A moving Control Volume

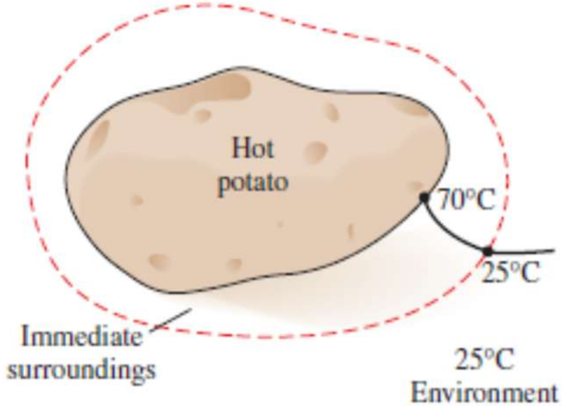
System, Immediate Surrounding, and Environment



As discussed, system is the part of the universe under observation for the thermodynamic analysis

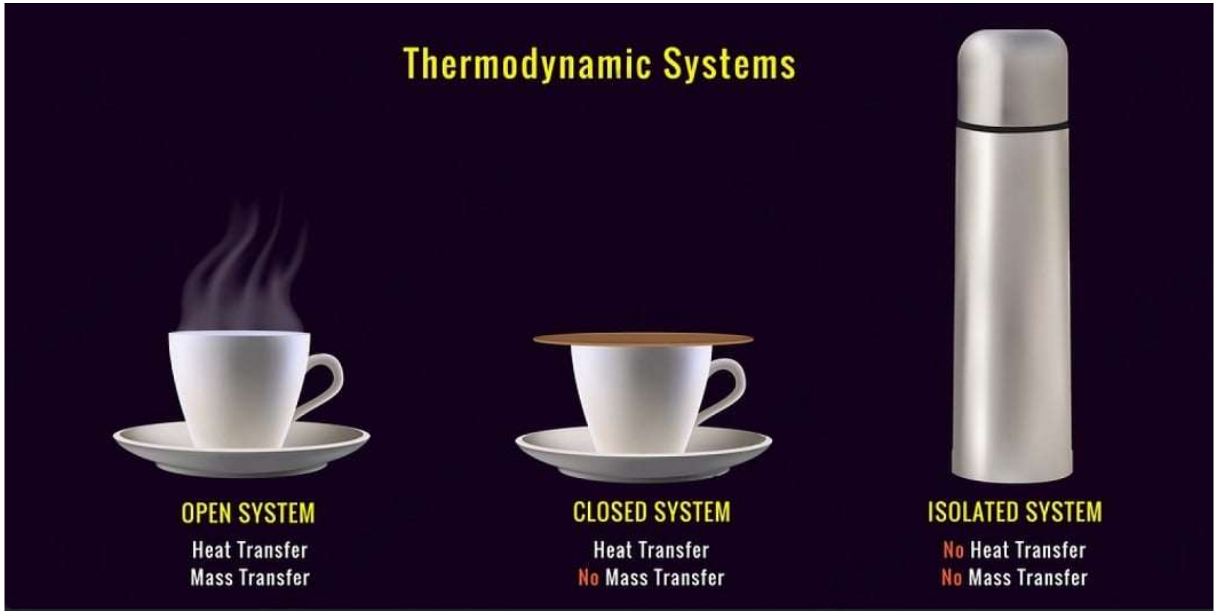
The part of the surrounding i.e. affected by the system due to energy/mass interactions is called immediate surrounding. e.g. If you stand on the back side of a window Ac up to some distance you can feel the warm air.

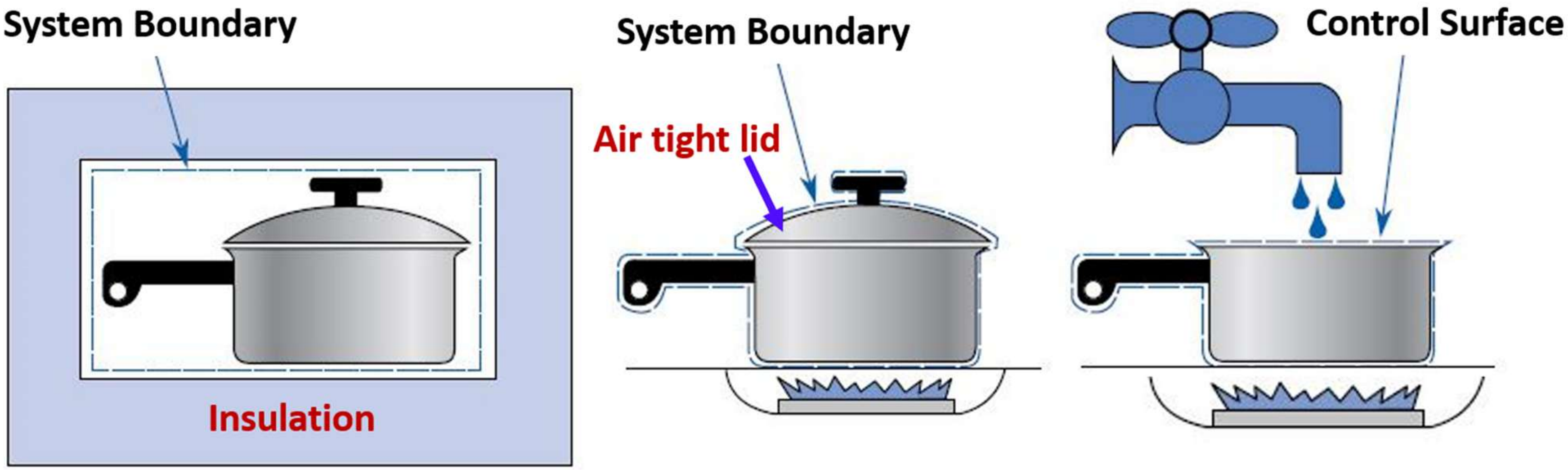
The concept of immediate surrounding will come when you have to consider an extended system to estimate the total entropy generation by the system (it is the system and immediate surrounding where the entropy get generated)



Everything outside the immediate surrounding is called environment. The environment behave as an isothermal reservoir and during any energy interaction the environment undergoes a reversible isothermal process. The system and surrounding (immediate surrounding +environment) together form an isolated system $\Delta E=0$. $\Delta S>=0$. The process undergone by the environment can be idealised as an internally reversible process as it is free from any entropy generation but its entropy changes due to transfer only.

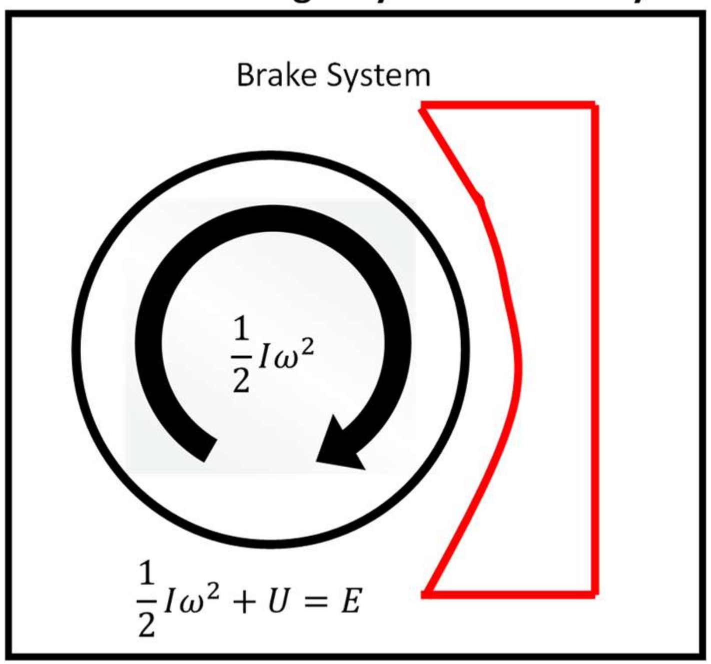
For an isolated system no mass and energy crosses the system boundary. Although no energy crosses the system boundary the different form of the stored energy readjust their selves till an equilibrium is reached.





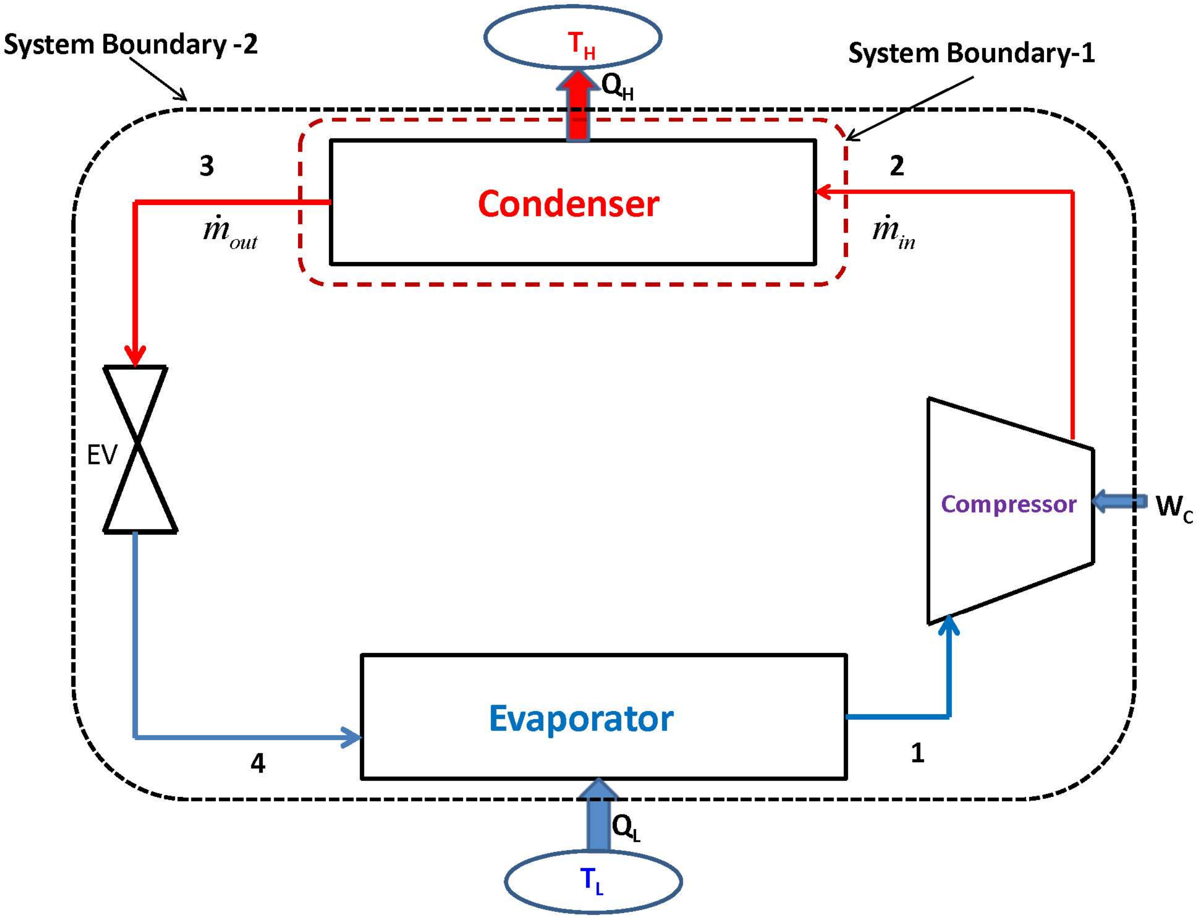
1) Isolated System 2) Closed System (Control Mass) 3) Open System (Control Volume)

Insulated & Rigid System Boundary



The important step in the study of thermodynamics is the identification of the system boundary as well as the system property

If you want to analyse the whole refrigeration system then the black line is the system boundary However, if you want to analyse the condenser (know the inlet and exit conditions of the condenser and the required energy /exergy interactions) then the red dotted line will be the system



Complete Refrigeration system (Closed system)

1st Law

$$COP = \frac{\dot{Q}_L}{\dot{W}_C} = \frac{\dot{Q}_L}{\dot{Q}_H - \dot{Q}_L}$$

2nd law

$$\dot{S}_{gen} = -\oint \frac{\delta \dot{Q}}{T} = -\left[\frac{\dot{Q}_L}{T_L} - \frac{\dot{Q}_H}{T_H}\right]$$

Condenser (Open system)

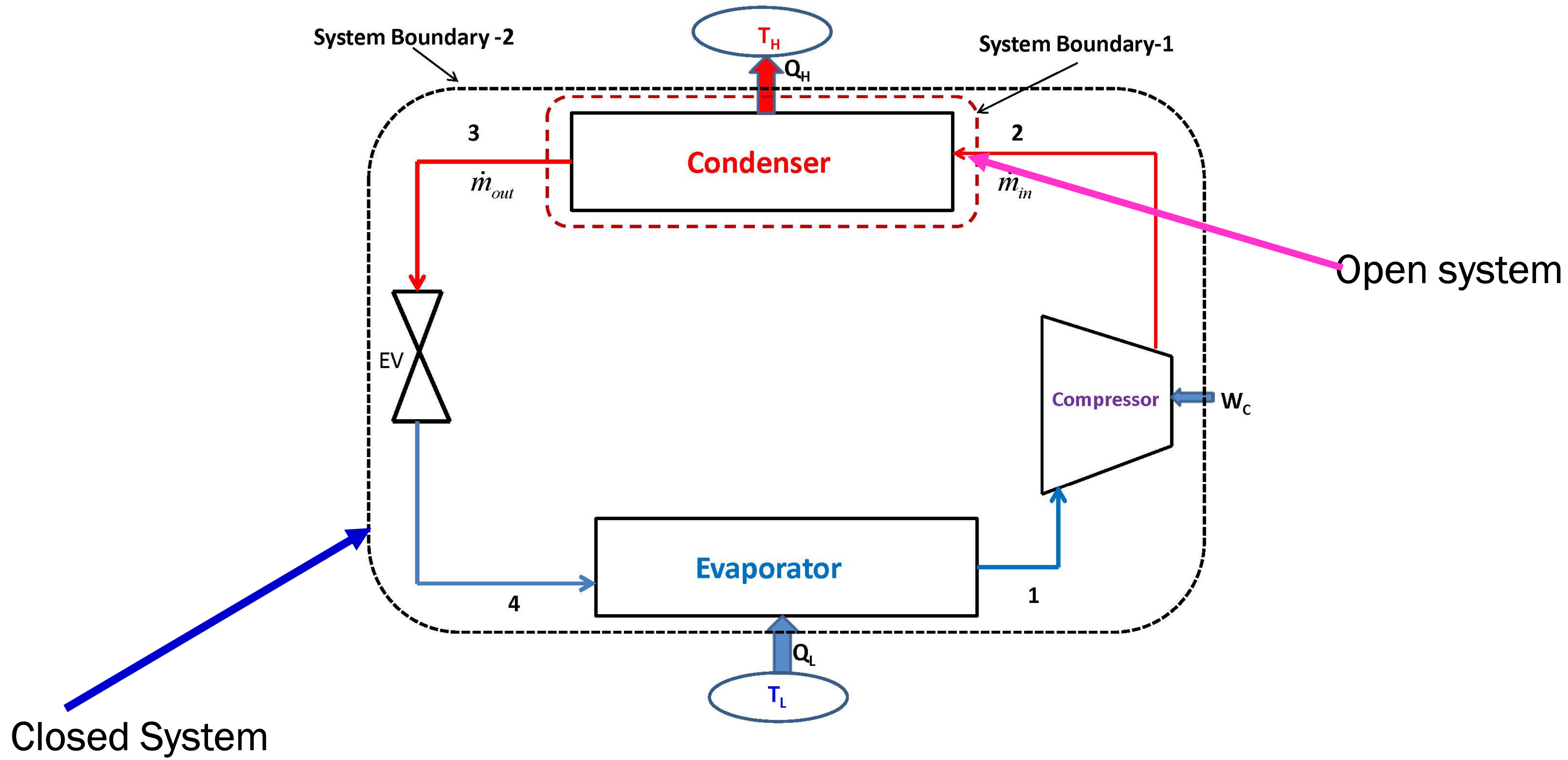
1st Law $\dot{Q}_H = \dot{m}(h_2 - h_3)$

2nd law

$$\dot{S}_{gen} = \dot{m}(s_3 - s_2) + \frac{\dot{Q}_H}{T_H}$$

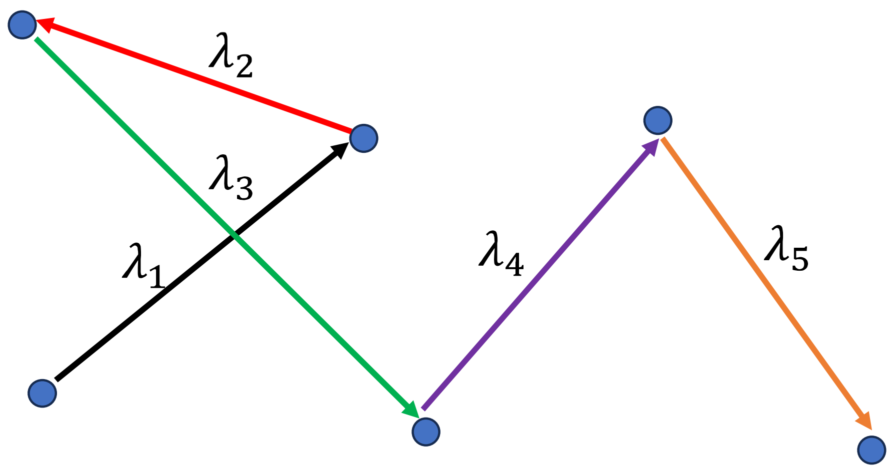
Note:

i. The system and its boundary have to be chosen by the analyst. In general, they are not specified in any problem.
ii. By changing the boundary an open system can become a closed system or vice versa.
iii. Choosing system boundary is often a matter of convenience.



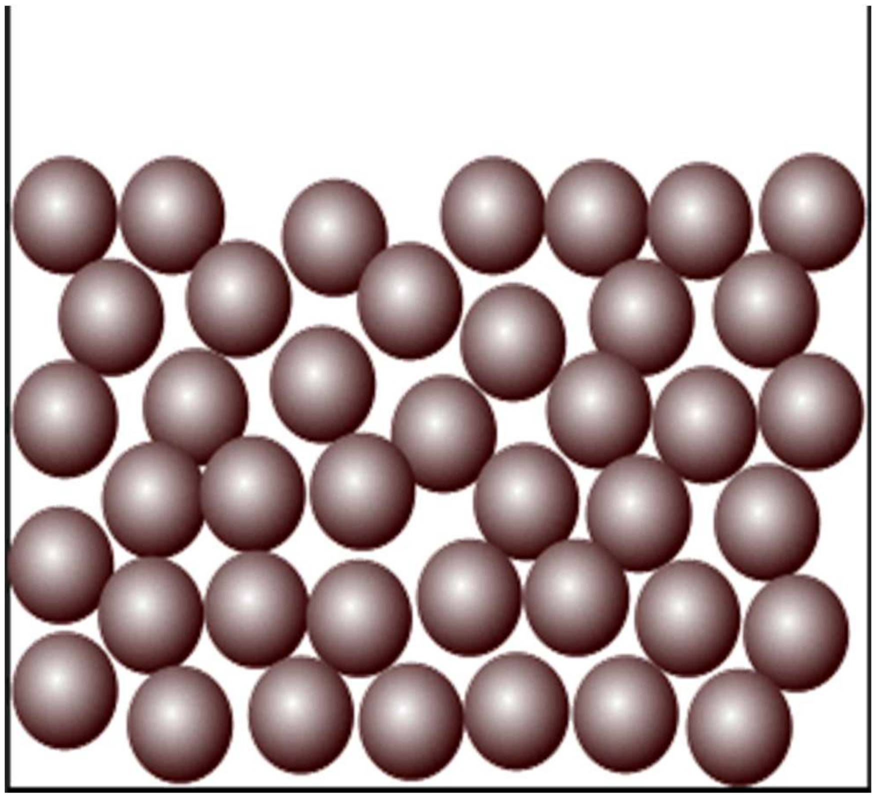
Continuum & Local Thermodynamic Equilibrium

- The continuum model assumes that the distribution of matter in a material is continuous although we know that it is not true.
- When the average inter atomic space in a solid or the average distance between the moving molecules of a liquid or the mean free path (that is the average distance traversed by the molecules of a gas between two consecutive collisions) of the molecules of a gas is small compared to characteristic length of the material, the continuum postulate yields accurate prediction. Additionally, the time interval over which a process takes place is very much longer than the average time between molecular collision.

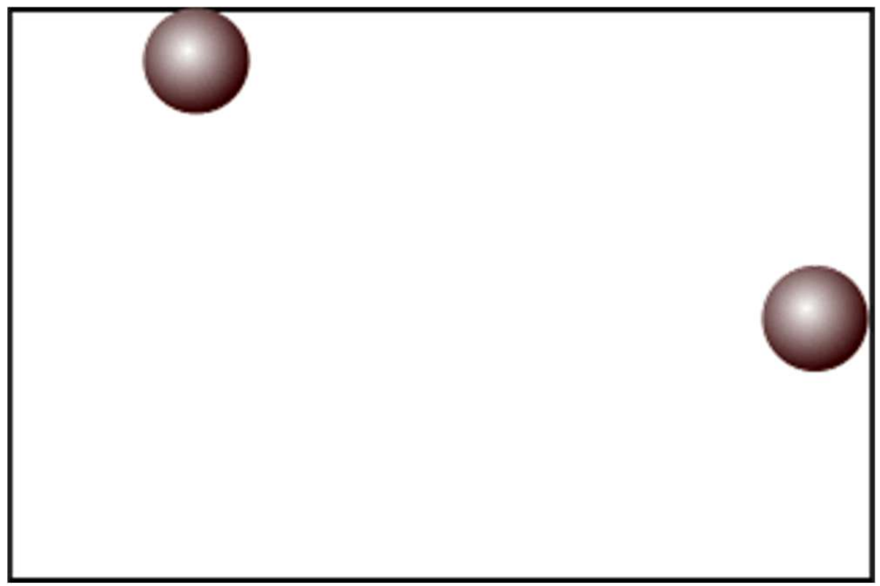


$$\lambda = \frac{\lambda_1 + \lambda_2 + \lambda_3 + \lambda_4 + \cdots \lambda_n}{n}$$

Gas contained in a closed box at very low pressure may not satisfy this condition ,i.e., the molecules of the gas will be far away from each other in this rarified condition and the mean free path will be comparable with that of the dimension of container



Continuum



Molecular

The study of thermodynamics, Heat Transfer, Fluid Mechanics depends on material properties such as specific volume, pressure, specific heat, thermal conductivity, viscosity etc. These familiar properties which we can quantify and measure are in fact manifestation of the molecular nature and activity of the material. All mater is composed of molecules which are in random motion and collision. In continuum model we ignore the characteristic of individual molecules and instead deal with the average or macroscopic effects. Thus, a continuum is assumed to be continuous in matter. This enables us to use the powerful tools of calculus to model or analyze the physical phenomena.

Criterion for validation of continuum assumption:

$$K_n = \frac{\lambda}{D_e} < 10^{-1}$$

In continuum model each point in the matter is occupied by a particle, and the particle is neither an atom nor a molecule, it is a hypothetical entity while atom and molecules are real entity. In the case of a liquid or gas a particle is a collection of large number of molecules occupying a region of $\Delta V'$ in the neighbourhood of appoint. In many situations the volume $\Delta V'$ is of the order of $10^{-9} mm^3$

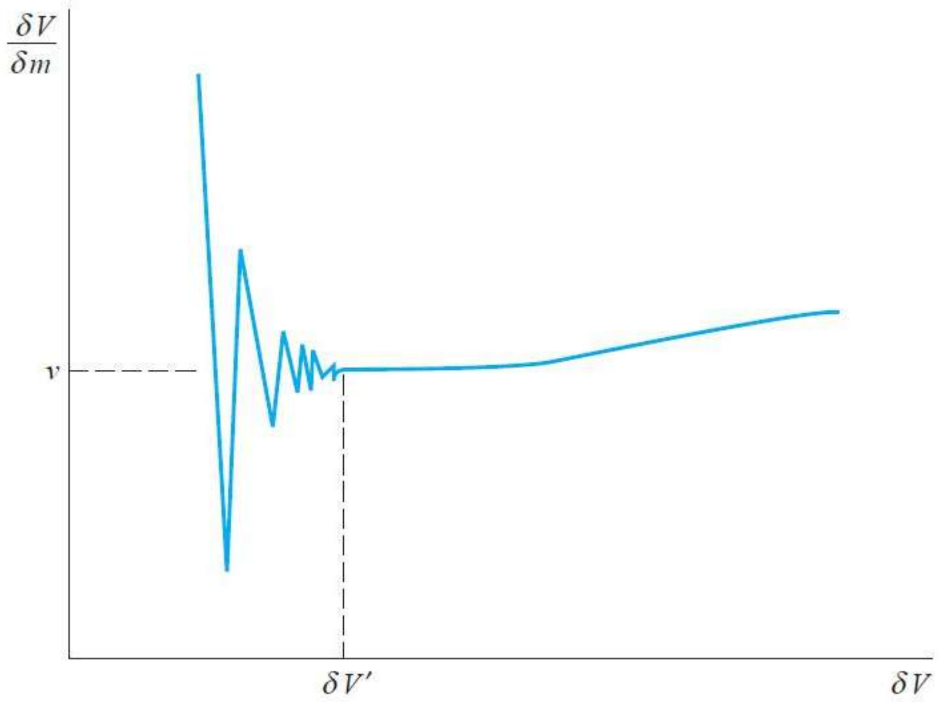
The specific volume of a system in a gravitational field may vary from point to point. For example, if the atmosphere is considered a system, the specific volume increases as the elevation increases. Therefore, the definition of specific volume involves the specific volume of a substance at a point in a system.

Consider a small volume δV of a system, and let the mass be designated δm. The specific volume is defined by the relation

$$v = \lim_{\delta V \to \delta V'} \frac{\delta V}{\delta m}$$

$10^{-9} mm^3$ of air at standard conditions contains approximately 3×10^7 molecules, which is sufficient to define a nearly constant density according to this equation

Where, $\Delta V'$ is the smallest volume for which the mass can be considered a continuum. Volumes smaller than this will lead to the recognition that mass is not evenly distributed in space but is concentrated in particles as molecules, atoms, electrons, etc. This is tentatively indicated in Fig. 1, where in the limit of a zero volume the specific volume may be infinite (the volume does not contain any mass) or very small (the volume is part of a nucleus).



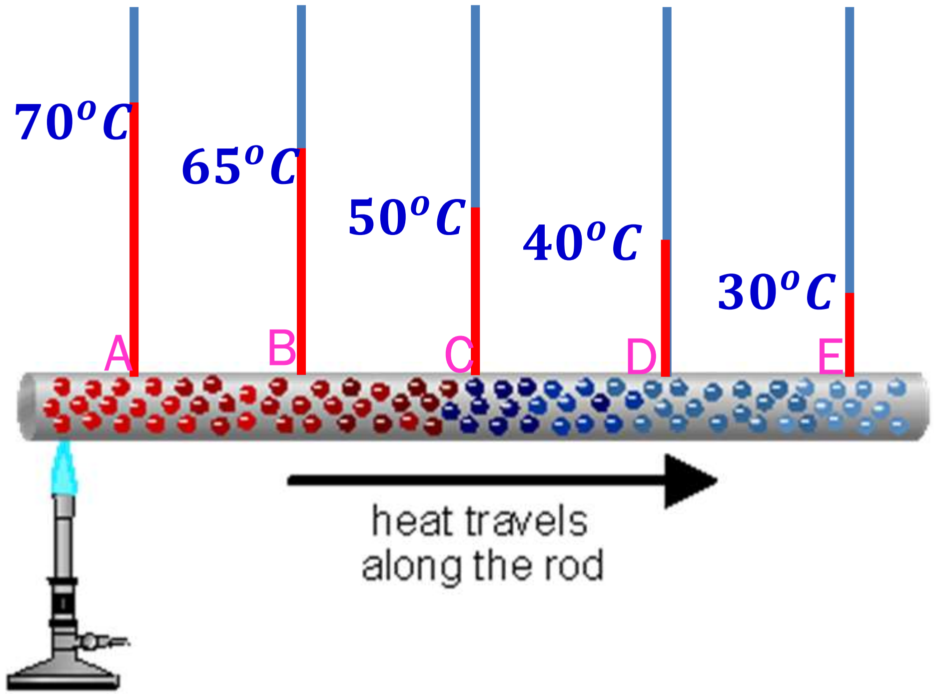
Similarly:

Pressure: $$P = \lim_{\delta A \to \delta A'} \frac{\delta F}{\delta A}$$

Heat Flux $$q'' = \lim_{\delta A_n \to \delta A'_n} \frac{\delta Q}{\delta A_n}$$

The local Thermodynamic Equilibrium Postulate

- There is no adequate definition for the thermodynamic properties of a system that is not in an equilibrium state .
- Some extension of classical equilibrium thermodynamics is necessary for us to be able to analyze non-equilibrium (or irreversible) process. We do this by subdividing a non equilibrium system into many small but finite volume element, each of which is larger than the local molecular mean free path so that the continuum hypothesis holds. We then assume that each of the small volume element is in local equilibrium with its surrounding elements.
- Thus, a nonequilibrium system can be broken down into a very large number of very small systems, each of which is at a different equilibrium state.



- The differential time quantity dt used in nonequilibrium thermodynamic analysis can not be allowed to go to zero as in normal calculus. We require that $dt > \sigma$, where, σ is the time it takes for one of the volume elements of the subdivided nonequilibrium state to an appropriate equilibrium state. This is analogous to not allowing the physical size of the element to be less than its molecular mean free path as required by the continuum hypothesis. The error incurred by these postulates is really quite small
- Thermodynamic equilibrium depends on the collision frequency of the molecules with an adjacent surface. When we are specifying temperature or any thermodynamic property at a point, we are indirectly assuming local thermodynamic equilibrium. This does not mean that the whole body is in thermodynamic equilibrium. At local thermodynamic equilibrium the fluid and the adjacent surface have the same velocity and temperature. This is called no velocity slip and no temperature jump condition.

The condition for local thermodynamic equilibrium

1. $K_n = \dfrac{\lambda}{D_e} < 10^{-3}$

2. $dt \gg \sigma$

Macroscopic vs. Microscopic Point of View

- A thermodynamic system can be studied either from a microscopic point of view or from a macroscopic point of view.
- Microscopic analysis is the analysis at the atomic level

Ex. A 1 cc amount of gas at STP consists of about 10^{20} molecules . If this is our system and we want to study it microscopically, then we have to specify the position (3 coordinates) and velocities (3 components) of each molecule⇒ a minimum of 6×10^{20} equations

- This is computationally impossible even with the latest super computer. To make this computationally feasible a statistical approach may be adapted (Statistical Thermodynamics)

- In statistical approach: A model of atom is considered. Using statistical and probability theory, average values for all molecules are obtained. Thus, number of equations are reduced which can be solved

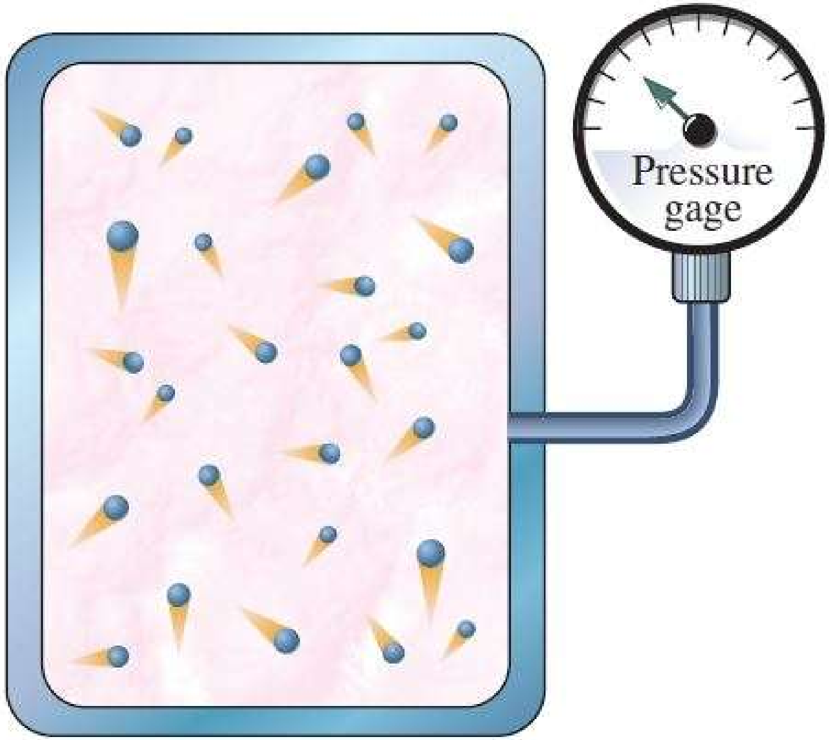
Using this approach:

a) Certain discontinuities in physical behavior such as super conductivity can be explained
b) Applications where continuum concept breaks down e.g. in case of rarefied gases as in outer space or high vacuum systems can be handled
c) Physical phenomena such as thermal conductivity and fluid viscosity, which are macroscopic properties but have origin at molecular level can be properly interpreted
d) Provides equation of state for non-measurable thermodynamic properties (e.g. internal energy, entropy) in terms of measurable thermodynamic properties (e.g. temperature, pressure , specific volume)

Macroscopic Analysis

- Macroscopic Analysis: is an alternative approach wherein the concept of continuum is applied to the system.
- In macroscopic analysis the average gross effect of innumerable molecules/atoms comprising the system, as perceived by our senses and that can be measured by instruments are considered.
- What we really perceive /measure is a time averaged influence of many molecule

- Ex: Pressure from macroscopic point of view is the time averaged force on a given area, which can be measured by a pressure gauge. From microscopic point of view this is a result of the rate of change of momentum of individual molecules colliding the wall of the container.



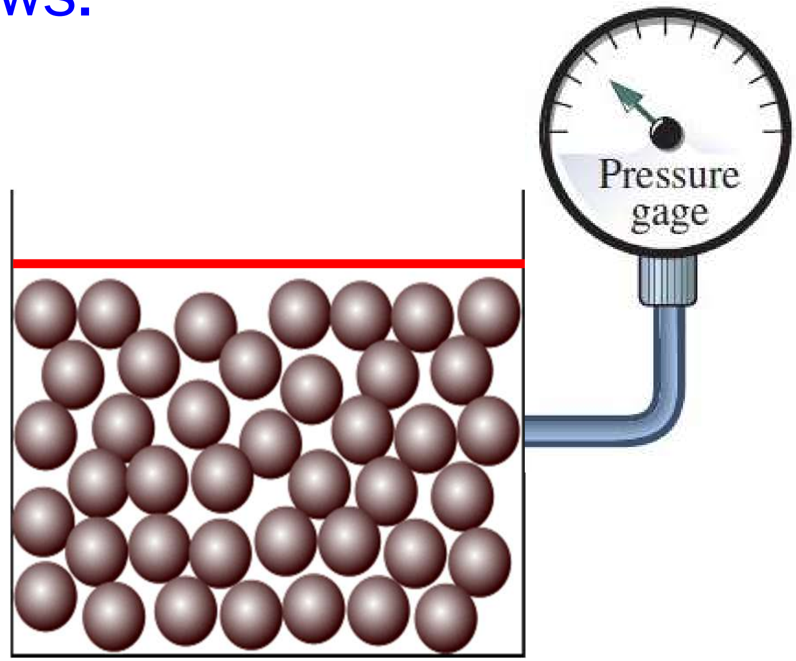
❖ The macroscopic observations are completely independent of our assumption regarding the nature of the matter

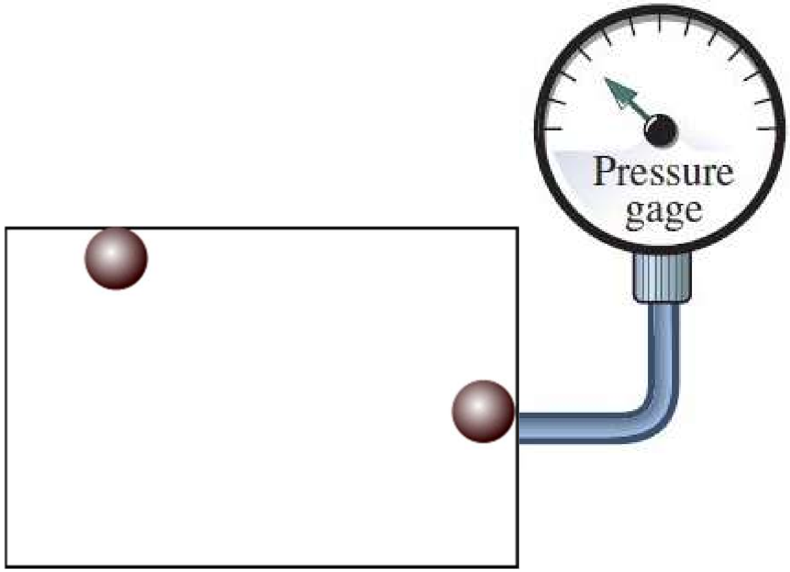
❖ The macroscopic analysis based on continuum hypothesis works very well as long as:

I. The dimensions of the systems being analysed are much larger than the dimensions of the molecules.

II. The time interval over which a process takes place is very much longer than the average time between molecular collision

❖ The Classical Thermodynamics is mainly based on Macroscopic views.





Pressure: we talk about pressure when dealing with fluids (liquids or gases) and stresses when dealing with solids

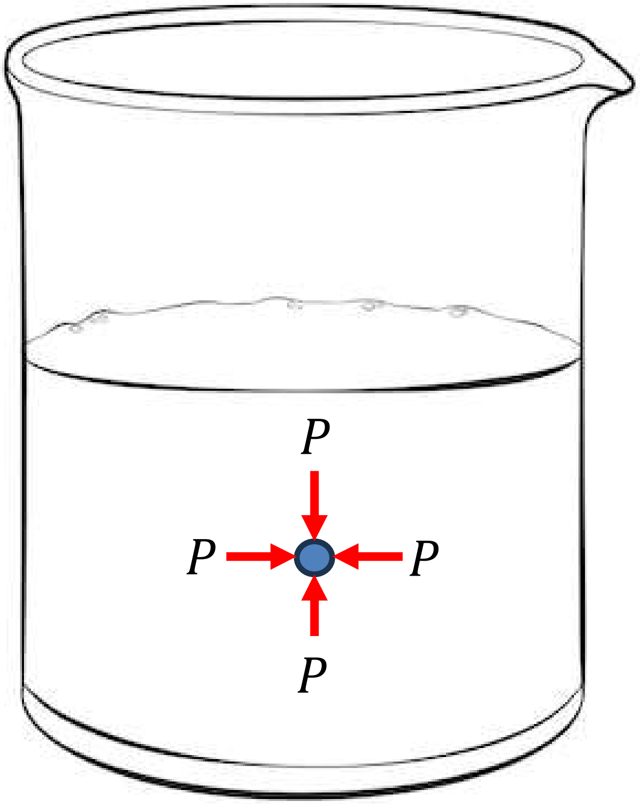
For fluids at rest, pressure is defined as the normal compressive force per unit area.

$$P = \lim_{\delta A \to \delta A'} \frac{\delta F_n}{\delta A}$$

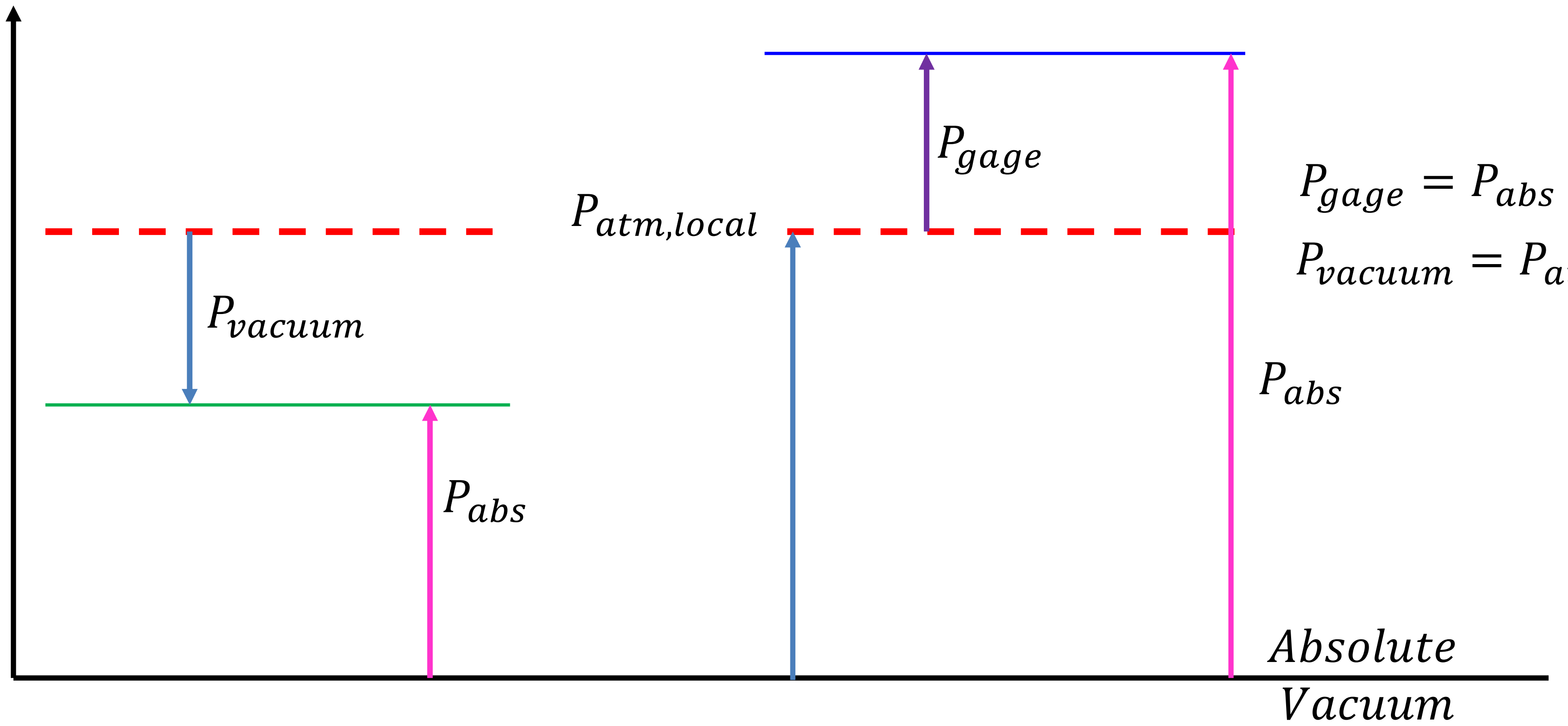
$\delta A'$ such that continuum concept holds good

For a fluid at rest, the pressure at a point is uniform in all directions when the fluid is at equilibrium.

The actual pressure at a given position called the absolute pressure, and it measured relative to absolute vacuu (i.e., absolute zero pressure).



Most pressure-measuring devices, however, are calibrated to read zero in the atmosphere and so they indicate the difference between the absolute pressure and the local atmospheric pressure. This difference is called the gage pressure. P_{gage} can be positive or negative, but pressures below atmospheric pressure are sometimes called vacuum pressures and are measured by vacuum gages that indicate the difference between the atmospheric pressure and the absolute pressure. Absolute, gage, and vacuum pressures are related to each other by

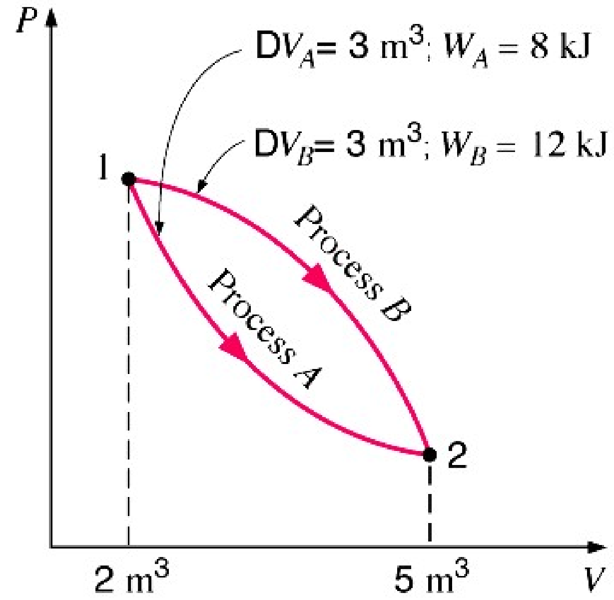


$$P_{gage} = P_{abs} - P_{atm,local}$$

$$P_{vacuum} = P_{atm,local} - P_{abs}$$

Important Definitions

- Property is any macroscopic characteristic or attribute of matter which can be evaluated quantitatively.
- It may be measurable (P, T, V, m) or calculated (U, H, S, G, A)
- Property is a point function and the change in property is independent of path followed. Hence the differential of any property is exact differential.



If P is a property such that p=f(T,v)

$$dp = \left(\frac{\partial p}{\partial T}\right)_v dT + \left(\frac{\partial p}{\partial v}\right)_T dv$$

- If dz is an exact differential of z=f (x ,y) then dz can be written as:

$$dz = Mdx + Ndy$$

Such that $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

- Another characteristics of the property is: $\oint d(any\, property) = 0$

Cyclic integral of the differential of any property is zero

Let R be defined as: $R = \int pdv + vdp$

State whether R is a property or not?

Now $dR = pdv + vdp = Mdx + Ndy$ Where $M = p;\ x = v;\ N = v;\ y = p$

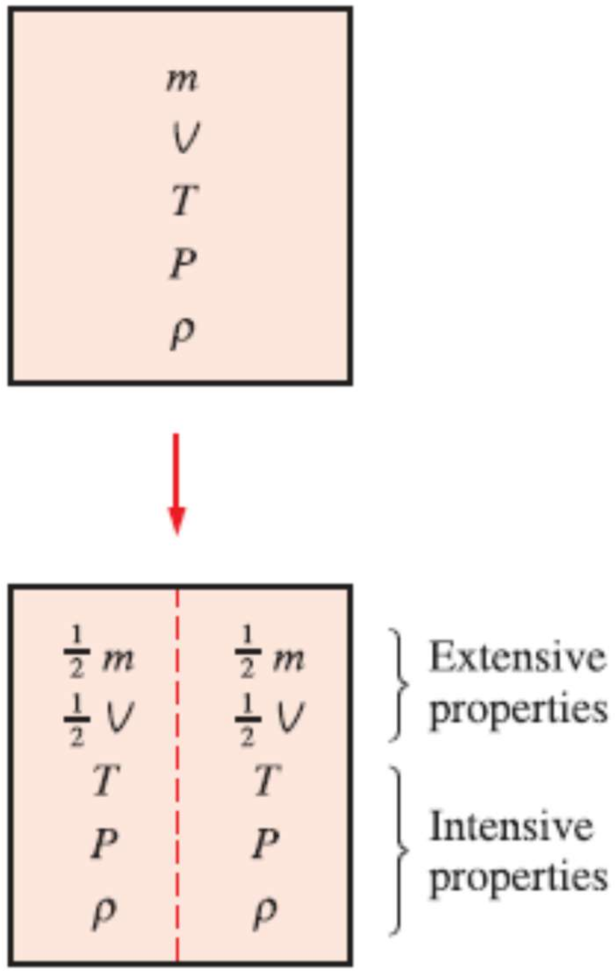
$$\frac{\partial M}{\partial y} = \frac{\partial p}{\partial p} = 1 = \frac{\partial N}{\partial x} = \frac{\partial v}{\partial v}$$

Hence the differential is exact, So "R" is property

If you check in other way

$$R = \int pdv + vdp = \int d(pv) = pv$$

- Thermodynamic properties can be intensive (independent of extent or size and mass of the system,(e.g. pressure, temperature, density) or extensive (depends on size & mass (e.g. mass ,volume). Extensive properties are additive in nature.



For a particular thermodynamics system, the differential of pressure is given by

$$dp = \frac{RdT}{v-b} - \left\{\frac{RT}{(v-b)^2} - \frac{2a}{v^3}\right\} dv$$

Where T is temperature, v is specific volume, a, b, R are constants

Find

$$dp = \left(\frac{\partial p}{\partial T}\right)_v dT + \left(\frac{\partial p}{\partial v}\right)_T dv$$

a) Where p is a property of the system or not

b) If it is a property find the relationship between p, v, T

$$dp = \frac{RdT}{v-b} - \left\{\frac{RT}{(v-b)^2} - \frac{2a}{v^3}\right\} dv \dots\dots (1)$$

$$dp = Mdx + Ndy \dots\dots (2)$$

$$M = \frac{R}{v-b} \quad N = -\left\{\frac{RT}{(v-b)^2} - \frac{2a}{v^3}\right\} \quad x = T \quad y = v$$

$$\frac{\partial M}{\partial y} = \frac{\partial \left(\frac{R}{v-b}\right)}{\partial v} = \frac{-R}{(v-b)^2}$$

$$\frac{\partial N}{\partial y} = \frac{\partial \left(-\left\{\frac{RT}{(v-b)^2} - \frac{2a}{v^3}\right\}\right)}{\partial T} = \frac{-R}{(v-b)^2}$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$ *Hence p is property*

$$dp = \frac{RdT}{v-b} - \left\{\frac{RT}{(v-b)^2} - \frac{2a}{v^3}\right\} dv \dots \dots (1)$$

$$dp = \left(\frac{\partial p}{\partial T}\right)_v dT + \left(\frac{\partial p}{\partial v}\right)_T dv$$

$$\left(\frac{\partial p}{\partial T}\right)_v = \frac{RdT}{v-b} \Longrightarrow p = \int \frac{R}{v-b}\bigg|_v dT = \frac{RT}{v-b} + f(v)$$

$$\left(\frac{\partial p}{\partial v}\right)_T = \frac{-RT}{(v-b)^2} + f'(v) = -\left\{\frac{RT}{(v-b)^2} - \frac{2a}{v^3}\right\}$$

$$f'(v) = \frac{df}{dv} = \frac{2a}{v^3}$$

$$f = \frac{-a}{v^2} + \mathrm{c}$$

$$\boldsymbol{p = \frac{RT}{v-b} + f(v) = \frac{RT}{v-b} - \frac{a}{v^2} + \mathbf{c}}$$